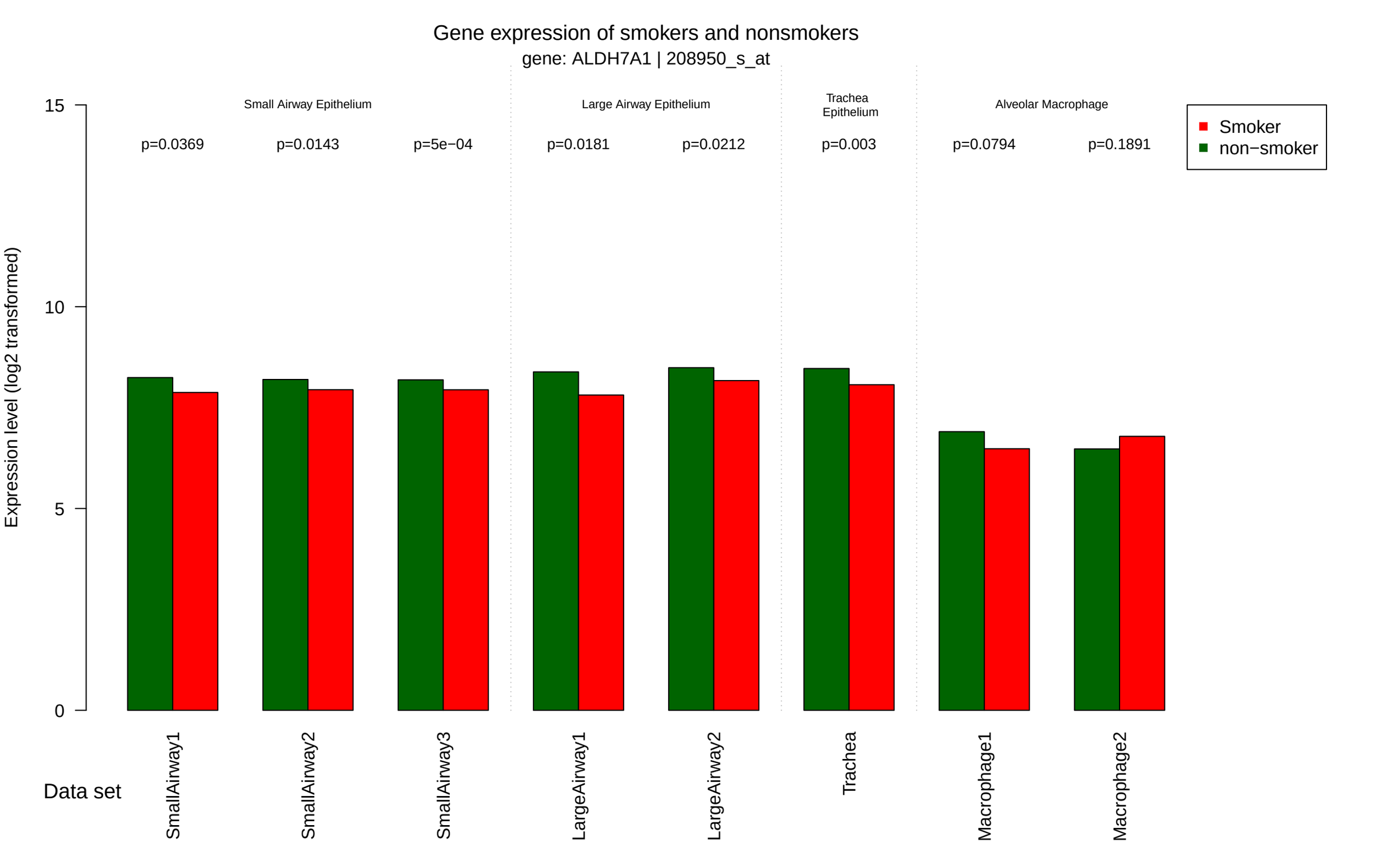
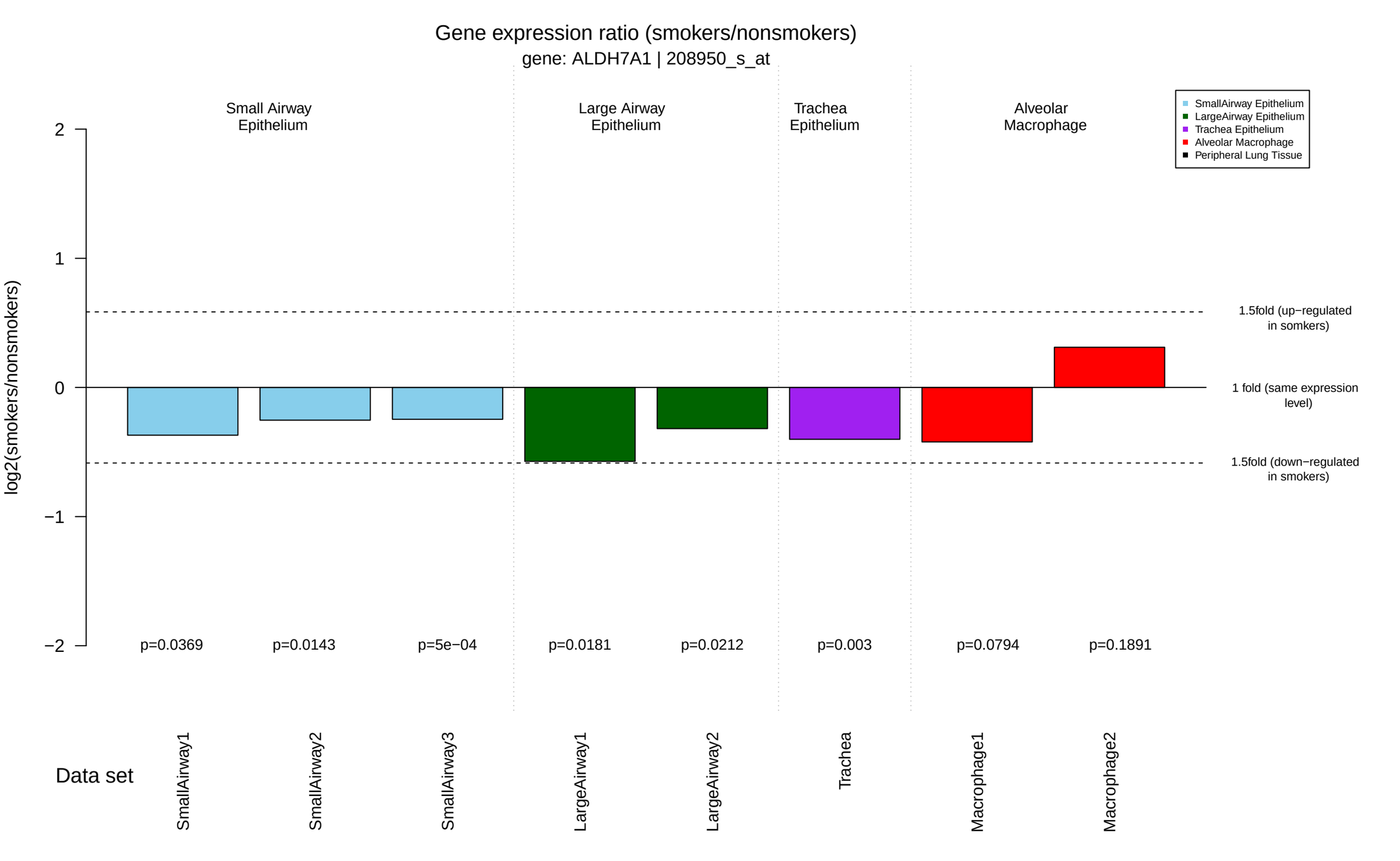
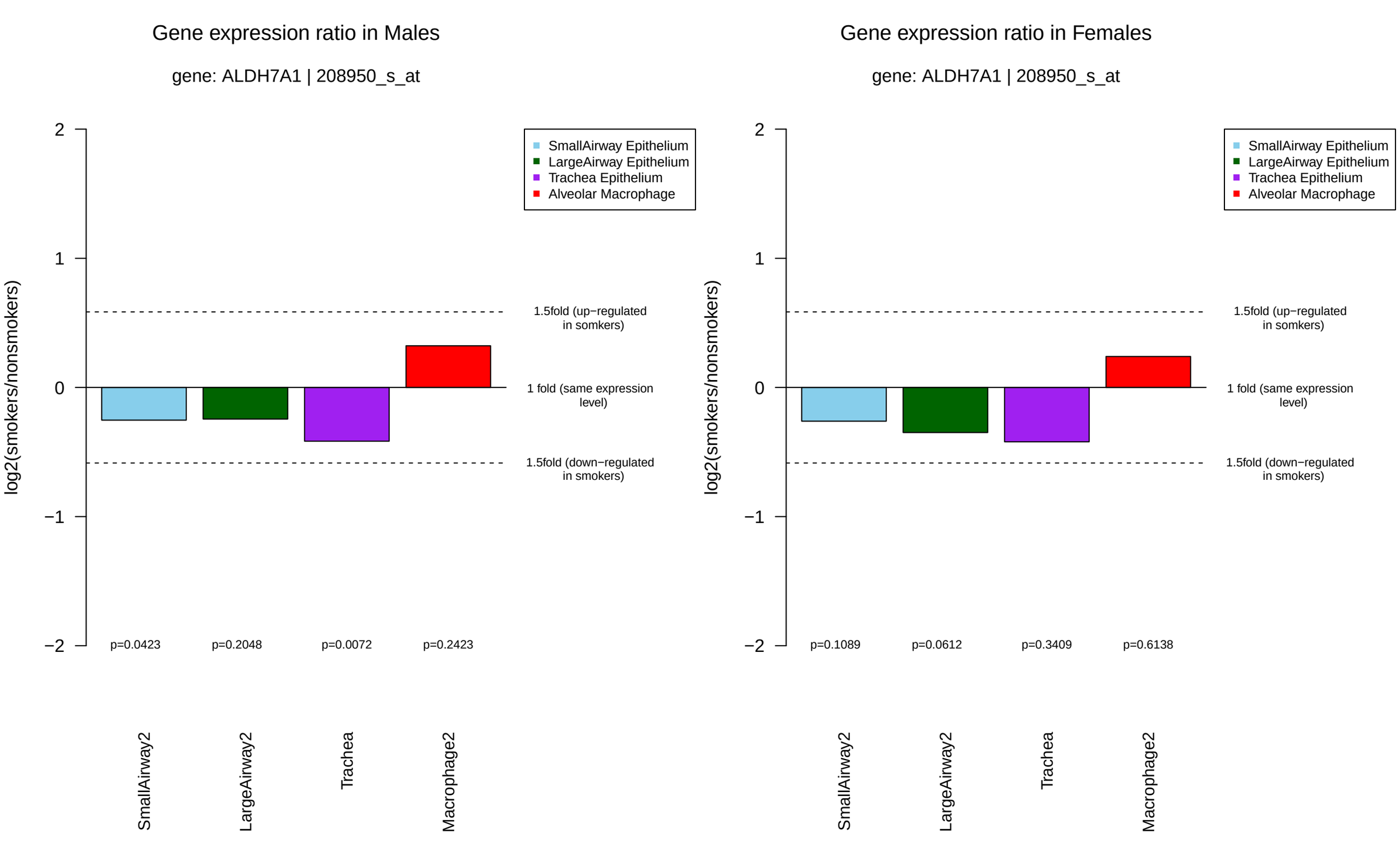
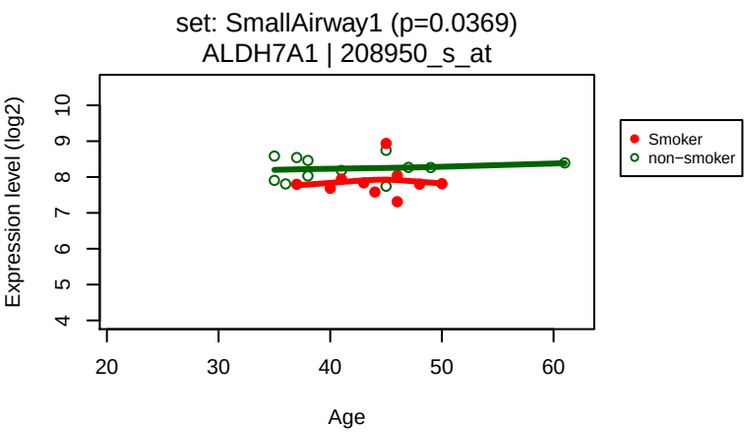
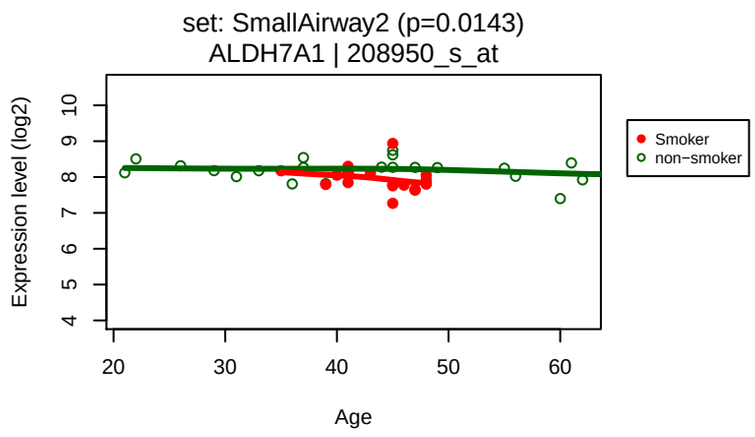
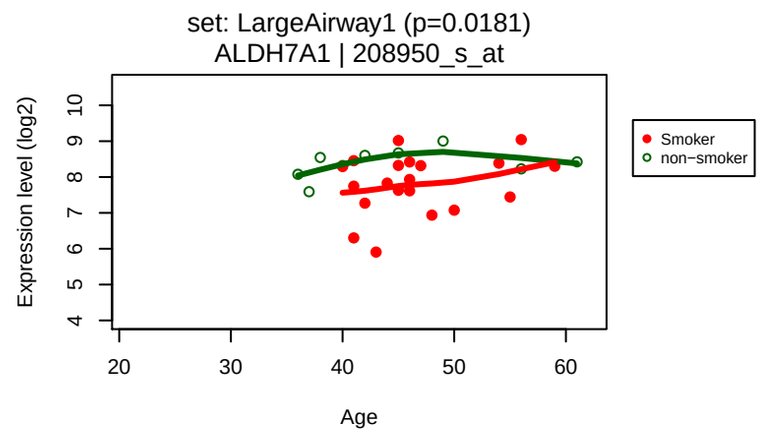

Gene expression of smokers and nonsmokers
gene: ALDH7A1 | 208950_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0369
p=0.0143
p=5e−04
p=0.0181
p=0.0212
p=0.003
p=0.0794
p=0.1891
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

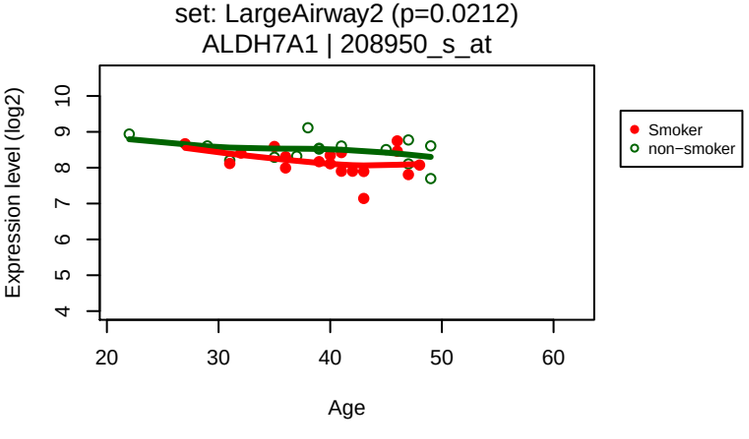

Gene expression ratio (smokers/nonsmokers)
gene: ALDH7A1 | 208950_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0369
p=0.0143
p=5e−04
p=0.0181
p=0.0212
p=0.003
p=0.0794
p=0.1891
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

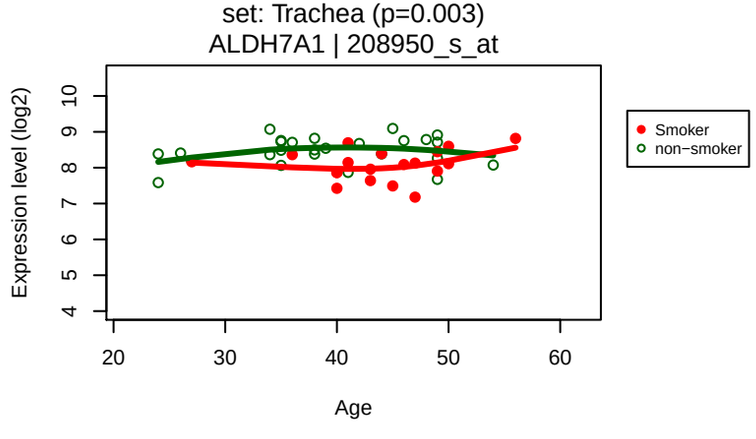

Gene expression ratio in Males
gene: ALDH7A1 | 208950_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0423
p=0.2048
p=0.0072
p=0.2423
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALDH7A1 | 208950_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1089
p=0.0612
p=0.3409
p=0.6138
SmallAirway2
LargeAirway2
Trachea
Macrophage2

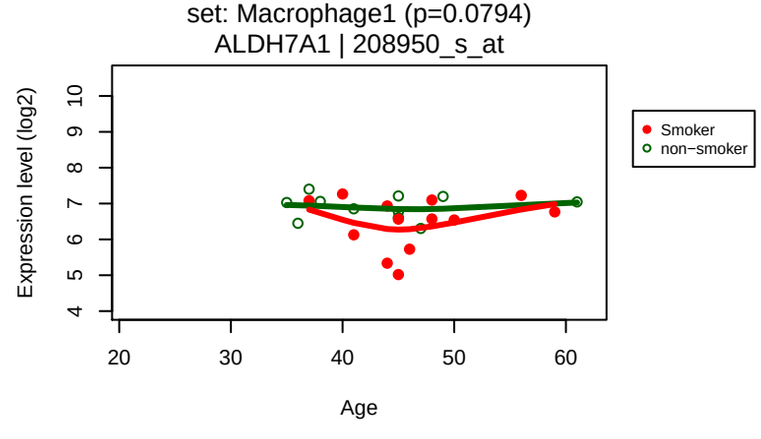

set: SmallAirway1 (p=0.0369)
ALDH7A1 | 208950_s_at
Expression level (log2)
Age
Smoker
non−smoker

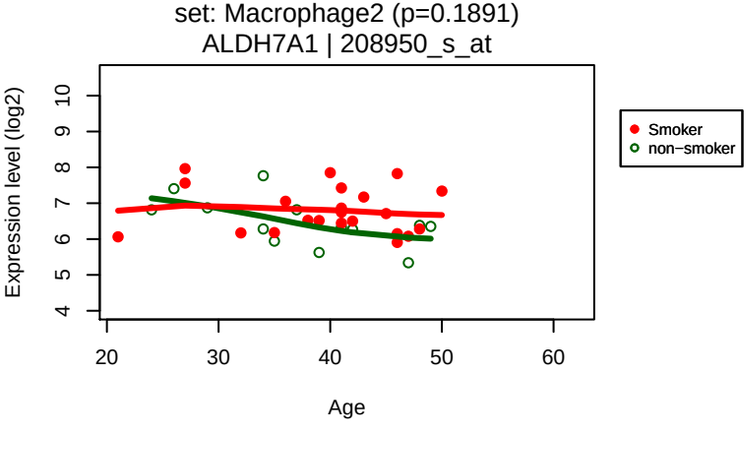

set: SmallAirway2 (p=0.0143)
ALDH7A1 | 208950_s_at
Expression level (log2)
Age
Smoker
non−smoker

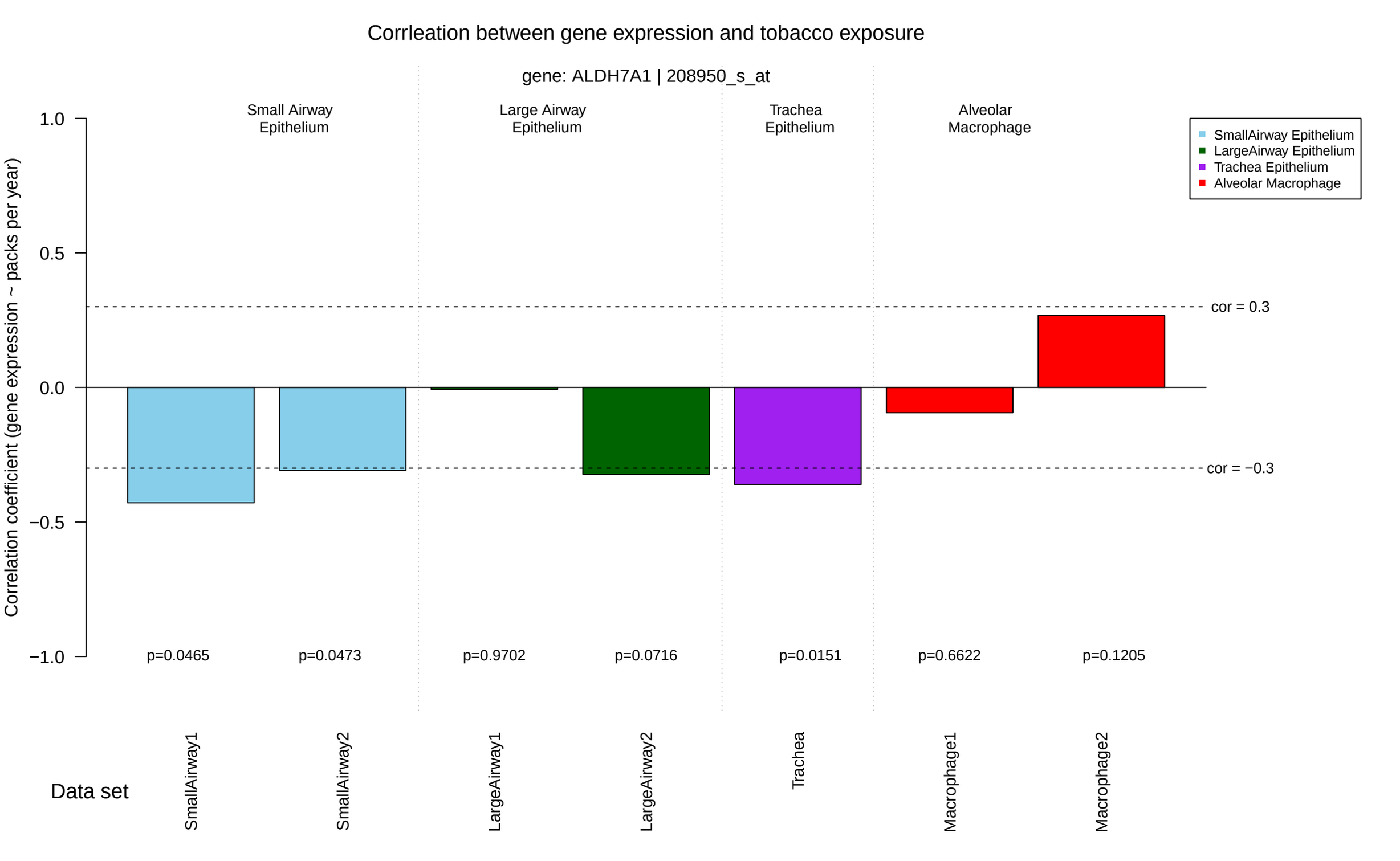

set: LargeAirway1 (p=0.0181)
ALDH7A1 | 208950_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0212)
ALDH7A1 | 208950_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.003)
ALDH7A1 | 208950_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0794)
ALDH7A1 | 208950_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1891)
ALDH7A1 | 208950_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALDH7A1 | 208950_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0465
p=0.0473
p=0.9702
p=0.0716
p=0.0151
p=0.6622
p=0.1205
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2